

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Bachelor of Technology (Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)****Course :Bachelor of Technology (Computer Engineering) Branch : Engineering and Technology****Semester : SEMESTER - 4****Subject Code & Name: BTCOC401 - DESIGN & ANALYSIS OF ALGORITHMS****Time : 3 Hours]****[Total Marks : 60****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)**12**

1	Which of the following is incorrect? Algorithms can be represented:			
	a) As program	b) As flowchart	c) As Pseudo code	As Syntax
	What is the result of the recurrences which fall under first case of Master's theorem			
	a) $T(n) = O(n^c \log a)$	b) $T(n) = O(n^c \log n)$	c) $T(n) = O(f(n))$	d) $T(n) = O(n^2)$
	What is the worst case time complexity of a quick sort algorithm?			
	a) $O(N)$	b) $O(N \log N)$	c) $O(N^2)$	d) $O(\log N)$
	The complexity of Binary search algorithm is			
	a) $O(n)$	b) $O(\log)$	c) $O(n^2)$	d) $O(n \log n)$
5	Backtracking algorithm is implemented by constructing a tree of choices called as?			
	a) State-space tree	b) State-chart tree	c) Node tree	d) Backtracking tree
6	The problem of finding a subset of positive integers whose sum is equal to a given positive integer is called as?			
	a) n- queen problem	b) sum of subset problem	c) knapsack problem	d) Hamiltonian cycle problem
	Fractional knapsack problem is solved most efficiently by which of the following algorithm?			
	a) Backtracking	b) Greedy algorithm	c) Dynamic programming	d) Divide and conquer
	Which of the following is true about Huffman Coding?			
	a. Huffman coding may become loss in some cases	Huffman Codes may not be optimal lossless codes in some cases	In Huffman coding, no code is prefix of any other code.	All of the above
	Which of the following problems should be solved using dynamic programming?			
	a) Merge sort	b) Binary search	c) Longest common subsequence	d) Quicksort
10	Consider the two matrices P and Q which are 10×20 and 20×30 matrices respectively. What is the number of multiplications required to multiply the two matrices?			
	a) 10×20	b) 20×30	c) 10×30	d) $10 \times 20 \times 30$
11	A problem which is both _____ and _____ is said to be NP complete.			
	a) NP, P	b) NP, NP hard	c) P, P complete	d) None of the mentioned

12	What does NP stands for in complexity classes theory?		
	a) Non polynomial	b) Non-deterministic polynomial	c) All of the mentioned
			d) None of the mentioned

Q2. Solve the following.

A) Solve the following Recurrence Relations using Masters Theorem.

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1. $T(n) = 4T(n/2) + n$
2. $T(n) = 9T(n/3) + n^2$

B) Solve the following Recurrence Relations using Recursive Tree.
 $T(n) = 2T(n/2) + cn$.

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Q3. Solve the following.

A) Multiply following two matrices using Strassen's matrix multiplication algorithm.

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$$\text{Matrix } A = \begin{bmatrix} 2 & 3 \\ 6 & 8 \end{bmatrix} \quad \text{Matrix } B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

B) Write an algorithm for merge sort, Apply merge sort on Following array.

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$$A = \{5, 1, 2, 6, 3, 7, 9, 4\}$$

Q4. Solve Any Two of the following.

A) Given $n=4$ weights $w = \{10, 20, 30, 40\}$ and $M=50$. Find all possible subsets for which $\text{sum} = M$ using sum of subsets.

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B) How 4- Queens problem is solved by backtracking approach? Explain with the help of state space tree.

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Solve the following 15-Puzzle Problem

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1	2	3	4
5	6	?	8
9	10	7	11
13	14	15	12

Q5. Solve Any Two of the following.

A) Apply Greedy approach to solve following Job Sequencing with Deadlines.

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Jobs	J1	J4	J3	J2
Profit	100	27	15	10
Deadline	2	2	2	1

B) Apply Greedy approach to solve following fractional knapsack problem if capacity of knapsack is 20 kg.
 Calculate maximum profit, $n=3, m=20$,
 Profit $-(P1, P2, P3) = (25, 24, 15)$
 Weight $-(W1, W2, W3) = (18, 15, 10)$

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Apply Huffman Coding algorithm on following data. Write the code for each character using Huffman Coding algorithm.

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Character	a	b	c	d	e	f
Frequency	45	13	12	16	9	5

Q6. Solve Any Two of the following.

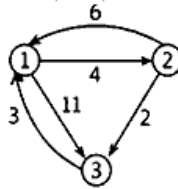
A) Determine Longest Common Subsequence between X and Y if

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$$X = \langle A, B, C, D \rangle \text{ and } Y = \langle A, E, D, F \rangle$$

B) Apply Floyd Warshall's algorithm on following graph.

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C) Explain complexity class P and complexity class NP.

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